

Honors Math 3

Unit 3 Polynomials

Day	Date	Lesson	Classwork	Homework					
M	2/22	Catapult Lab - Gather Data Benchmark Review	Warm-up Review	100	92	84	76	68	60
Tu	2/23	Benchmark #1 HW: Laws of Exponents	Benchmark Exponents	100	92	84	76	68	60
W	2/24	Polynomial Vocabulary Operating with Polynomials	Warm-up p. 1-2	100	92	84	76	68	60
Th	2/25	Dividing Polynomials Remainder/Factor Theorem	Warm-up p. 3-6	100	92	84	76	68	60
F	2/26	Review Polynomials Quiz	Warm-up p. 1-7	100	92	84	76	68	60
M	2/29	Graphing Polynomials	Warm-up p. 8-13	100	92	84	76	68	60
Tu	3/1	Solving Polynomial Equations	Warm-up p. 14-17	100	92	84	76	68	60
W	3/2	Writing Equations from Zeros	Warm-up p. 18-19	100	92	84	76	68	60
Th	3/3	Review	Warm-up Finish Packet/ Review	100	92	84	76	68	60
F	3/4	Test	Do Not Panic - You know this!	HW- Take first set of notes for Unit 4 Simplify/Multiply/Divide Rational Expressions					

Unit 3 Warm-ups

Date: _____

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Unit 3 Warm-ups

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Unit 3 Warm-ups

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Name _____

Write each polynomial in standard form. Name each polynomial by its number of terms (monomial, binomial, trinomial, polynomial) and state the degree.

		Standard Form	Name	Degree	L.C.
1.	$3x - 2x^8$				
2.	$-x - 7x^3 - 3x^2 + 5$				
3.	$25x$				
4.	$-2x^2 + 6x^3 - 7x$				
5.	$-7x^3 - 4x + 6$				

*Add or subtract the polynomials.

6. $(3x^2 - 5x + 7) + (-x^2 + 2x - 10)$

6. _____

7. $(8x + 6x^2 + 2) + (9x^2 - 7x - 6)$

7. _____

8. $(7x^2 + 5x + 6) - (8x^2 - 2x + 4)$

8. _____

9. $(y^2 - 7y) - (-4y^2 + 3y - 1)$

9. _____

10. $(2a^3 + 9a - 2) - (a^2 + 4a - 7)$

10. _____

*Multiply the polynomials.

11. $4x^3(-x^2 + 6x - 2)$

11. _____

12. $(5x - 2)(-3x + 4)$

12. _____

13. $(5x - 2)^2$

13. _____

14. $(x + 3)(x - 3)$

14. _____

15. $(x - 4)(2x^2 + 3x - 5)$

15. _____

*Divide each polynomial by the monomial. Make sure your answer is in simplest form!

16. $(6x^2 - 18x - 9) \div 9x$

16. _____

17. $(y^2 + 6y + 2) \div 3y$

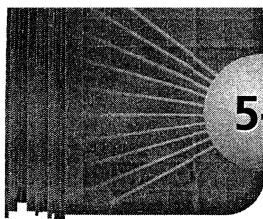
17. _____

18. $(10z^4 + 5z^3 - z^2) \div 5z^3$

18. _____

19. $(4a^2b^2 - 8a^3b^2 + 6ab) \div 2ab^2$

19. _____



5-2 Practice

Polynomials

Determine whether each expression is a polynomial. If it is a polynomial, state the degree of the polynomial.

1. $5x^3 + 2xy^4 + 6xy$

2. $-\frac{4}{3}ac - a^5d^3$

3. $\frac{12m^8n^9}{(m-n)^2}$

4. $25x^3z - x\sqrt{78}$

5. $6c^{-2} + c - 1$

6. $\frac{5}{r} + \frac{6}{s}$

Simplify.

7. $(3n^2 + 1) + (8n^2 - 8)$

8. $(6w - 11w^2) - (4 + 7w^2)$

9. $(-6n - 13n^2) + (-3n + 9n^2)$

10. $(8x^2 - 3x) - (4x^2 + 5x - 3)$

11. $(5m^2 - 2mp - 6p^2) - (-3m^2 + 5mp + p^2)$

12. $(2x^2 - xy + y^2) + (-3x^2 + 4xy + 3y^2)$

13. $(5t - 7) + (2t^2 + 3t + 12)$

14. $(u - 4) - (6 + 3u^2 - 4u)$

15. $-9(y^2 - 7w)$

16. $-9r^4y^2(-3ry^7 + 2r^3y^4 - 8r^{10})$

17. $-6a^2w(a^3w - aw^4)$

18. $5a^2w^3(a^2w^6 - 3a^4w^2 + 9aw^6)$

19. $2x^2(x^2 + xy - 2y^2)$

20. $-\frac{3}{5}ab^3d^2(-5ab^2d^5 - 5ab)$

21. $(v^2 - 6)(v^2 + 4)$

22. $(7a + 9y)(2a - y)$

23. $(y - 8)^2$

24. $(x^2 + 5y)^2$

25. $(5x + 4w)(5x - 4w)$

26. $(2n^4 - 3)(2n^4 + 3)$

27. $(w + 2s)(w^2 - 2ws + 4s^2)$

28. $(x + y)(x^2 - 3xy + 2y^2)$

29. BANKING Terry invests \$1500 in two mutual funds. The first year, one fund grows 3.8% and the other grows 6%. Write a polynomial to represent the amount Terry's \$1500 grows to in that year if x represents the amount he invested in the fund with the lesser growth rate.

30. GEOMETRY The area of the base of a rectangular box measures $2x^2 + 4x - 3$ square units. The height of the box measures x units. Find a polynomial expression for the volume of the box.

5-3

Practice

Dividing Polynomials

Simplify.

1. $\frac{15r^{10} - 5r^8 + 40r^2}{5r^4}$

2. $\frac{6k^2m - 12k^3m^2 + 9m^3}{2km^2}$

3. $(-30x^3y + 12x^2y^2 - 18x^2y) \div (-6x^2y)$

4. $(-6w^3z^4 - 3w^2z^5 + 4w + 5z) \div (2w^2z)$

5. $(4a^3 - 8a^2 + a^2)(4a)^{-1}$

6. $(28d^3k^2 + d^2k^2 - 4dk^2)(4dk^2)^{-1}$

7. $\frac{f^2 + 7f + 10}{f + 2}$

8. $\frac{2x^2 + 3x - 14}{x - 2}$

9. $(a^3 - 64) \div (a - 4)$

10. $(b^3 + 27) \div (b + 3)$

11. $\frac{2x^3 + 6x + 152}{x + 4}$

12. $\frac{2x^3 + 4x - 6}{x + 3}$

13. $(3w^3 + 7w^2 - 4w + 3) \div (w + 3)$

14. $(6y^4 + 15y^3 - 28y - 6) \div (y + 2)$

15. $(x^4 - 3x^3 - 11x^2 + 3x + 10) \div (x - 5)$

16. $(3m^5 + m - 1) \div (m + 1)$

17. $(x^4 - 3x^3 + 5x - 6)(x + 2)^{-1}$

18. $(6y^2 - 5y - 15)(2y + 3)^{-1}$

19. $\frac{4x^2 - 2x + 6}{2x - 3}$

20. $\frac{6x^2 - x - 7}{3x + 1}$

21. $(2r^3 + 5r^2 - 2r - 15) \div (2r - 3)$

22. $(6t^3 + 5t^2 - 2t + 1) \div (3t + 1)$

23. $\frac{4p^4 - 17p^2 + 14p - 3}{2p - 3}$

24. $\frac{2h^4 - h^3 + h^2 + h - 3}{h^2 - 1}$

25. **GEOMETRY** The area of a rectangle is $2x^2 - 11x + 15$ square feet. The length of the rectangle is $2x - 5$ feet. What is the width of the rectangle?

26. **GEOMETRY** The area of a triangle is $15x^4 + 3x^3 + 4x^2 - x - 3$ square meters. The length of the base of the triangle is $6x^2 - 2$ meters. What is the height of the triangle?

Dividing Polynomials: Long & Synthetic Division

Date _____ Period _____

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Divide two ways: a) Long division and b) Synthetic division

1) $(n^3 + n^2 - 11n + 18) \div (n - 1)$

2) $(m^3 + 6m^2 + 9m - 5) \div (m + 1)$

Divide using synthetic division.

3) $(x^3 - 6x^2 - 6x + 9) \div (x + 1)$

4) $(8n^3 - 9n^2 - 13n - 6) \div (n - 2)$

5) $(3r^3 + 34r^2 + 89r + 75) \div (r + 8)$

6) $(6b^3 + 28b^2 + 25b - 21) \div (b + 3)$

Using synthetic division, state if the given binomial is a factor of the given polynomial.

7) $(8r^3 - 55r^2 + 44r - 12) \div (r - 6)$

8) $(n^3 - 9n - 3) \div (n + 3)$

9) $(b^3 + 7b^2 - 2) \div (b + 7)$

10) $(2x^3 + 28x^2 + 86x + 60) \div (x + 10)$

11) $(m^3 - 2m^2 - 11m - 6) \div (m + 2)$

12) $(v^3 - 10v^2 + 15v + 20) \div (v - 3)$

Using long division, state if the given binomial is a factor of the given polynomial.

13) $(7a^3 - 56a^2 - 105a - 51) \div (7a + 7)$

14) $(7x^3 - 33x^2 + 20x) \div (7x - 5)$

15) $(10n^3 + 43n^2 + 45n - 56) \div (10n - 7)$

16) $(7k^3 + 33k^2 - 22k - 21) \div (7k + 5)$

State if the given binomial is a factor of the given polynomial. Choose either long division or synthetic division.

17) $(10p^3 + 28p^2 + 14p + 4) \div (10p - 2)$

18) $(40x^3 + 106x^2 + 26x - 5) \div (10x + 4)$

19) $(8n^3 + 3n^2 - 3) \div (8n + 3)$

20) $(10b^3 + 74b^2 - 8b - 24) \div (10b - 6)$

Extra: If you finish early, choose a method to divide, make sure to practice both methods.

21) $(10n^3 - 37n^2 - 112n - 25) \div (10n + 3)$

22) $(3a^3 + 17a^2 + 19a) \div (3a + 2)$

23) $(5x^3 - 22x^2 + 20x + 29) \div (5x + 3)$

24) $(6r^3 + 4r^2 + 3) \div (6r + 4)$

25) $(36x^3 - 4x^2 - 9x) \div (9x - 1)$

26) $(2x^3 + 7x^2 - 18x - 22) \div (2x - 5)$

27) $(48a^3 + 20a^2 - 72a - 31) \div (6a + 7)$

28) $(18v^3 - 54v^2 + 46v - 50) \div (3v - 7)$

The Remainder Theorem

Date_____ Period____

Evaluate each function at the given value.

1) $f(x) = -x^3 + 6x - 7$ at $x = 2$

2) $f(x) = x^3 + x^2 - 5x - 6$ at $x = 2$

3) $f(a) = a^3 + 3a^2 + 2a + 8$ at $a = -3$

4) $f(a) = a^3 + 5a^2 + 10a + 12$ at $a = -2$

5) $f(a) = a^4 + 3a^3 - 17a^2 + 2a - 7$ at $a = 3$

6) $f(x) = x^5 - 47x^3 - 16x^2 + 8x + 52$ at $x = 7$

State if the given binomial is a factor of the given polynomial.

7) $(k^3 - k^2 - k - 2) \div (k - 2)$

8) $(b^4 - 8b^3 - b^2 + 62b - 34) \div (b - 7)$

9) $(n^4 + 9n^3 + 14n^2 + 50n + 9) \div (n + 8)$

10) $(p^4 + 6p^3 + 11p^2 + 29p - 13) \div (p + 5)$

11) $(p^4 - 8p^3 + 10p^2 + 2p + 4) \div (p - 2)$

12) $(n^5 - 25n^3 - 7n^2 - 37n - 18) \div (n + 5)$

13) $(x^5 + 6x^4 - 3x^2 - 22x - 29) \div (x + 6)$

14) $(n^4 + 10n^3 + 21n^2 + 6n - 8) \div (n + 2)$

Divide.

15) $(p^4 + 5p^3 - 11p^2 - 25p + 29) \div (p + 6)$

16) $(8k^3 - 66k^2 + 14k + 8) \div (k - 8)$

17) $(x^4 + 11x^3 + 33x^2 + 24x + 32) \div (x + 6)$

18) $(6v^3 + 42v^2 - 50v - 20) \div (v + 8)$

19) $(6b^4 + 53b^3 + 32b^2 - 61b + 19) \div (b + 8)$

20) $(4n^3 - 9n^2 + 9n + 3) \div (n - 1)$

21) $(6a^3 + 20a^2 - 15a + 9) \div (a + 4)$

22) $(n^4 - 6n^3 - 10n^2 + 20n + 15) \div (n + 2)$

23) $(p^5 + 8p^4 + 2p^2 + 19p + 16) \div (p + 8)$

24) $(x^4 - 2x^3 - 16x^2 + 28x + 9) \div (x - 4)$

25) $(r^5 + 6r^4 - 13r^3 - 5r^2 - 8r + 14) \div (r - 2)$

26) $(8v^5 + 32v^4 + 5v + 20) \div (v + 4)$

Notes: Solving Cubics

Textbook 6.5, 6.6

Yesterday, we factored cubics and quartics to find solutions to polynomial equations. However, not all polynomials can be factored. Today we will learn how to solve polynomial equations when factoring is not an option.

The **Rational Root Theorem** states that any factors of a polynomial equation can be found by $\frac{p}{q}$, where p are the factors of the _____ and q are the factors of the _____.

Using the Rational Root Theorem to Find Factors

Examples: Find the *rational roots* of the following equations.

A) $x^3 + x^2 - 3x - 3 = 0$

B) $x^3 - 4x^2 - 2x + 8 = 0$

Once you have used the Rational Root Theorem to find any rational roots, you can write this solution as a factor. For example, $x = 1$ can be written as $(x - 1)$. To find the remaining factors, we are going to divide the original polynomial by our factor using synthetic division.

Examples: Find *all* the roots of the following equations.

C) $2x^3 - x^2 - 2x + 1 = 0$

D) $x^3 - 2x^2 - 5x + 10 = 0$

You Try: Find all roots for $3x^3 + x^2 - x + 1 = 0$

Notes: Polynomial Functions

Textbook 6.1, 6.2

Classifying Polynomials

The degree of a polynomial is the largest degree of any term of the polynomial.

Degree	Name Using Degree	Polynomial Example
0	constant	6
1		
2		
3		
4		
5		

Example 1: Given the polynomial,
 $y = -7x^3 - 2x^2 + 3x + 2$,
 identify the following:

Cubic Term: _____

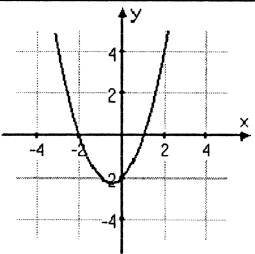
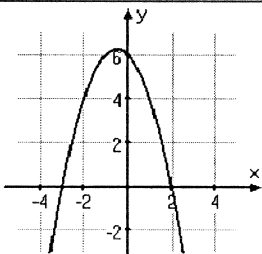
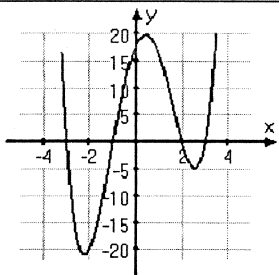
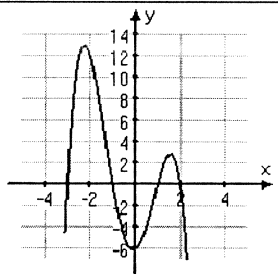
Quadratic Term: _____

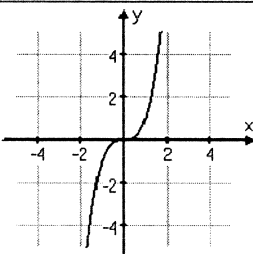
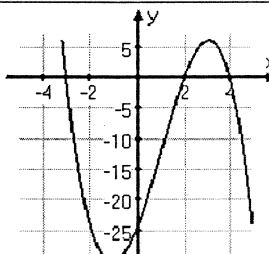
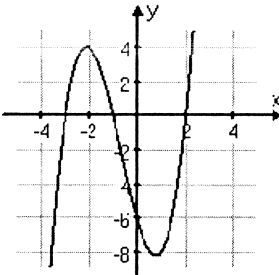
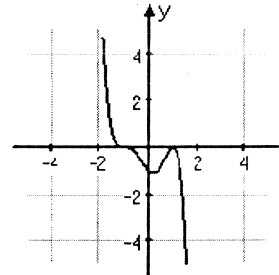
Linear Term: _____

Constant Term: _____

The coefficient in front of the largest degree is often referred to as the **leading coefficient**. What is the leading coefficient in Example 1?

The **end behavior** of a graph describes the far left and the far right portions of the graph. Examine the two tables below...

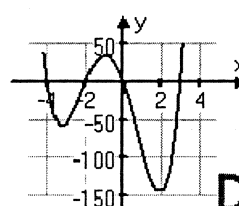
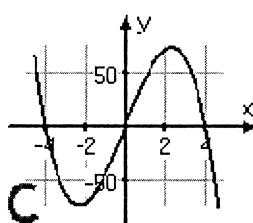
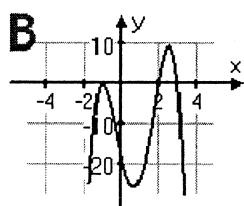
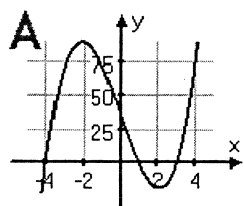
Even Degree Polynomials	
Positive Leading Coefficient	Negative Leading Coefficient
	
	

Odd Degree Polynomials	
Positive Leading Coefficient	Negative Leading Coefficient
	
	

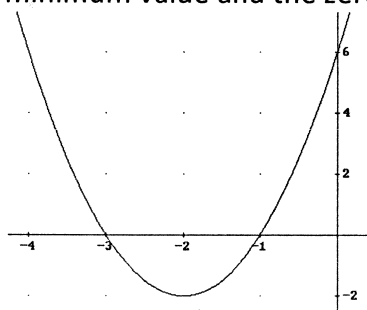
What do you notice about the end behavior in even degree polynomials vs. odd degree polynomials?

What difference does having a positive or negative leading coefficient have on the polynomial?

Example 2: Which of the following could be the graph of a polynomial whose leading term is $-3x^4$?



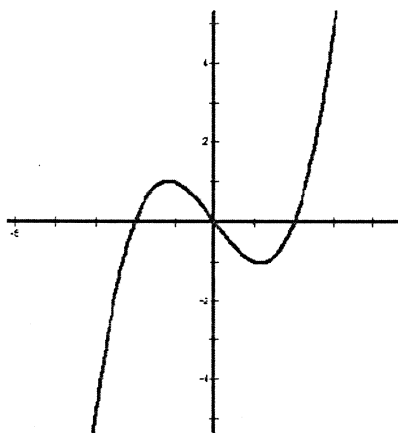
Last unit, we learned that a quadratic function has a degree of 2. A quadratic function has either a maximum value or a minimum value located at the vertex of the parabola. In the parabola below, list the minimum value and the zeros of the function.



Minimum: _____

Zeros: _____

A cubic function has a degree of 3. The graph of a cubic function looks like the picture below and contains a relative maximum and a relative minimum value. A relative maximum is the greatest value of the points in a region of the graph. Similarly, the relative minimum is the least y-value among nearby points on a graph. Find the relative min, relative max and zeros in the function below.



Relative Minimum: _____

Relative Maximum: _____

Zeros: _____

The **Corollary to the Fundamental Theorem of Algebra** says that every polynomial in one variable of degree $n > 0$ has exactly n , not necessarily distinct, real or complex zeros. So, what this means is that a polynomial of degree 3, will have 3 zeros. A polynomial of degree 4 will have 4 zeros, etc.

Example 3: Find the relative max and min for each function.

a) $y = 3x^3 - 8x^2 + 5$

b) $y = -2x^3 - 7x^2 + 3$

Relative Min: _____

Relative Min: _____

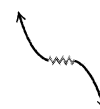
Relative Max: _____

Relative Max: _____

Worksheet 4.1: Roughly Graphing Polynomials (End Behavior; Zeros and Their Multiplicity)

Leading Term Test:

If n is even, and $a_n > 0$: If n is even, and $a_n < 0$: If n is odd, and $a_n > 0$: If n is odd, and $a_n < 0$:



Multiplicity of zeros: If the multiplicity of the zero is odd, then the graph passes through the x -axis at that zero. If the multiplicity is even, the graph 'bounces' there at the x -axis.

Equation: $y = x^2 - 9 = (x + 3)(x - 3)$

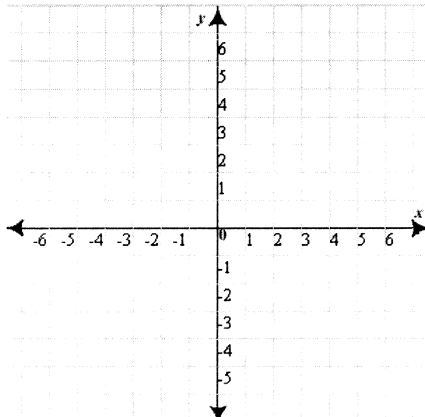
Zeros (multiplicity):

Is the degree even or odd?

Sign of the leading coefficient:

Thus the end behavior will be:

x	y



Equation: $y = x^2 - 6x + 9 = (x - 3)^2$

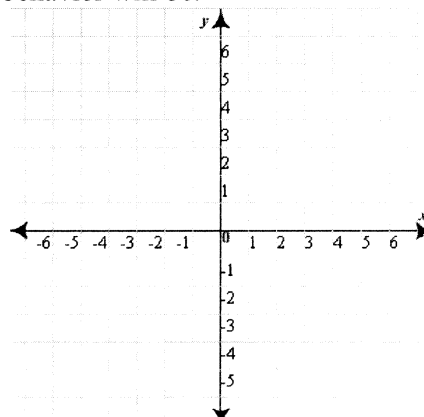
Zeros (multiplicity):

Is the degree even or odd?

Sign of the leading coefficient:

Thus the end behavior will be:

x	y



Equation: $y = (x + 3)^2(x - 3)$

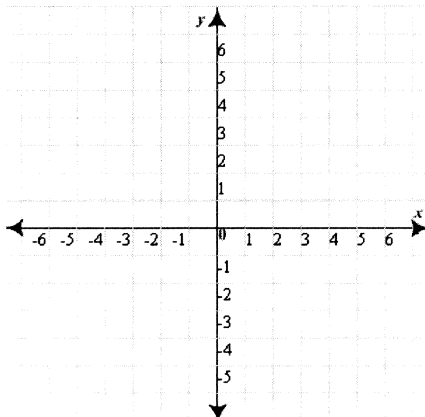
Zeros (multiplicity):

Is the degree even or odd?

Sign of the leading coefficient:

Thus the end behavior will be:

x	y



Equation: $y = x(x + 2)(x + 1)^2(x - 2)^2$

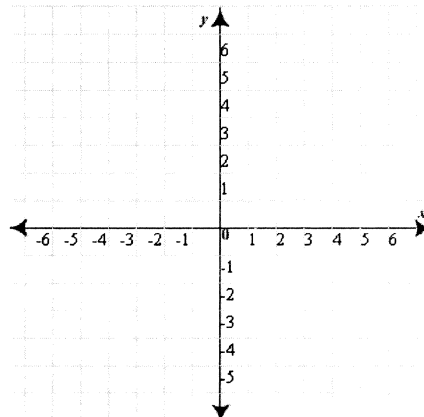
Zeros (multiplicity):

Is the degree even or odd?

Sign of the leading coefficient:

Thus the end behavior will be:

x	y



Worksheet 4.1: Classifying Polynomials

$$f(x) = x - 5x^2 + 7x^3$$

What kind of polynomial is $f(x)$?

What degree is $f(x)$?

What is the leading term of $f(x)$?

What is the leading coefficient of $f(x)$?

$$g(x) = -6x - 5$$

What kind of polynomial is $g(x)$?

What degree is $g(x)$?

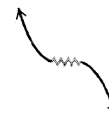
What is the leading term of $g(x)$?

What is the leading coefficient of $g(x)$?

Worksheet 4.1: Roughly Graphing Polynomials (Using End Behavior and Zeros of Multiplicity 1)

If n is even, and $a_n > 0$: If n is even, and $a_n < 0$: If n is odd, and $a_n > 0$: If n is odd, and $a_n < 0$:

Leading
Term Test:



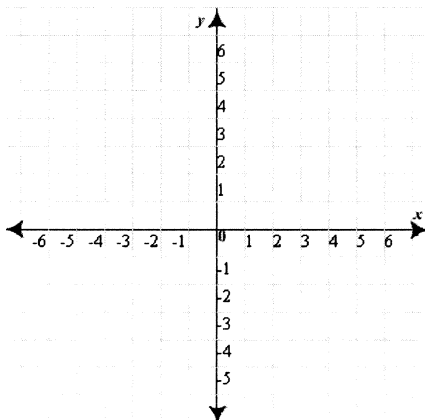
$$y = x^3 + 2x^2 - 11x - 12 = (x + 4)(x + 1)(x - 3)$$

Is the degree even or odd?

Sign of the leading coefficient:

Thus the end behavior will be:

x	y



The zeros are:

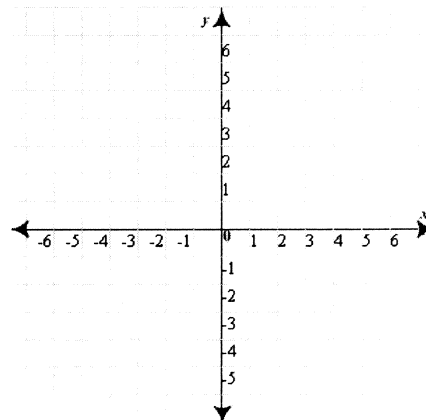
$$y = x^4 - 5x^2 + 4 = (x - 2)(x + 2)(x - 1)(x + 1)$$

Is the degree even or odd?

Sign of the leading coefficient:

Thus the end behavior will be:

x	y



The zeros are:

LESSON 6.9

NAME _____

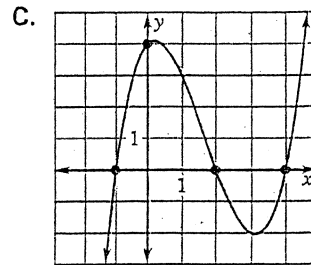
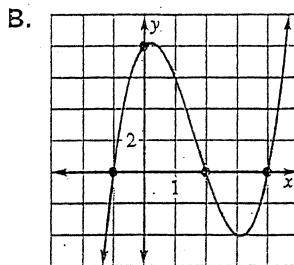
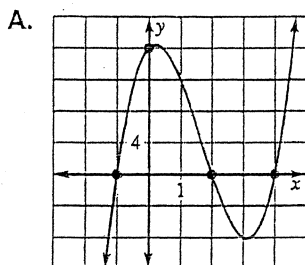
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Practice A

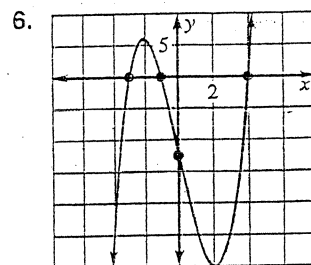
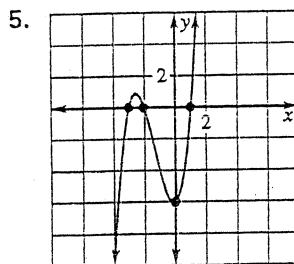
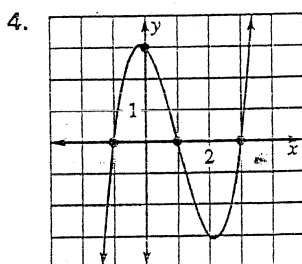
For use with pages 380–386

Match the cubic function with its graph.

1. $f(x) = (x + 1)(x - 2)(x - 4)$ 2. $f(x) = 2(x + 1)(x - 2)(x - 4)$ 3. $f(x) = \frac{1}{2}(x + 1)(x - 2)(x - 4)$

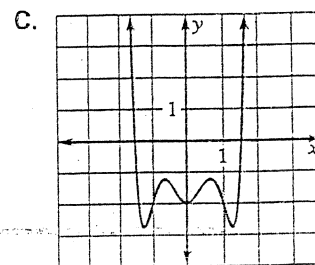
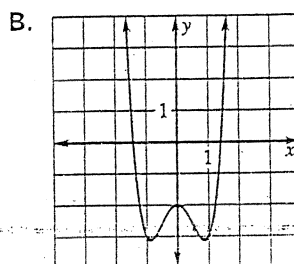
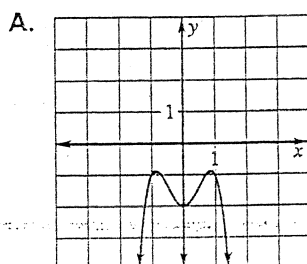


Write the cubic function whose graph is shown.



Match the graph with its function.

7. $f(x) = 2x^4 - 3x^2 - 2$ 8. $f(x) = 2x^6 - 6x^4 + 4x^2 - 2$ 9. $f(x) = -2x^4 + 3x^2 - 2$



Determine the x-intercepts of the graph of the function.

10. $f(x) = (x + 3)(x - 2)(x - 5)$ 11. $f(x) = (x + 4)(x - 6)(x - 8)$ 12. $f(x) = (x + 3)^2(x - 2)$
13. $f(x) = (x + 5)(x + 1)(x - 7)$ 14. $f(x) = (x + 6)^3(x + 2)$ 15. $f(x) = (x - 8)^5$

16. **Multiple Choice** Which of the following is the factorization of $8x^3 - 125$?

- (A) $(2x - 5)(4x^2 + 10x + 25)$
(B) $(2x - 5)(4x^2 - 10x + 25)$
(C) $(2x + 5)(4x^2 - 10x - 25)$
(D) $(2x + 5)(4x^2 + 10x - 25)$
(E) $(2x - 5)(4x^2 - 10x - 25)$

17. **Multiple Choice** Which of the following is the factorization of $x^3 - 5x^2 - 16x + 80$?

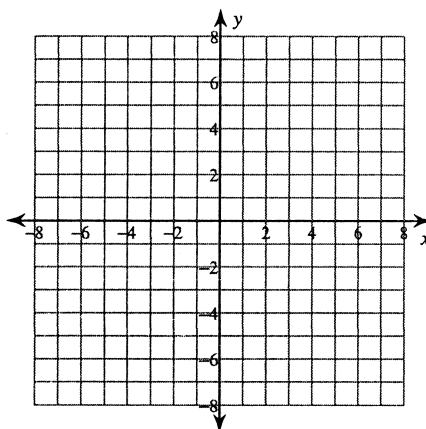
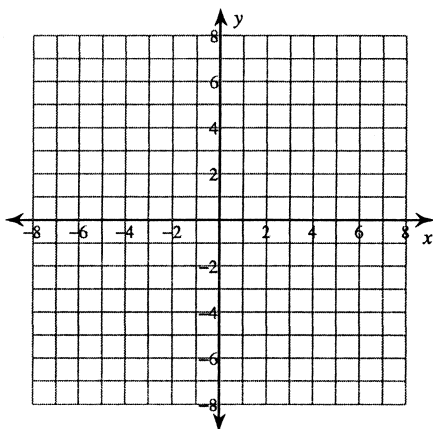
- (A) $(x - 4)(x + 4)(x + 5)$
(B) $(x - 4)^2(x + 5)$
(C) $(x - 4)(x + 5)^2$
(D) $(x - 4)(x + 4)(x - 5)$
(E) $(x + 4)^2(x + 5)$

Sketch the graph of each function. Approximate each real zero to the nearest tenth. Approximate the relative minima and relative maxima to the nearest tenth. Describe the end behavior and state

the Domain and Range.

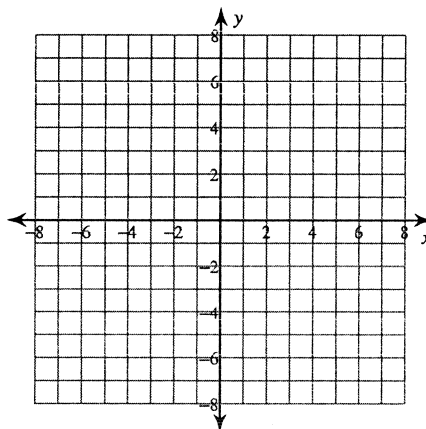
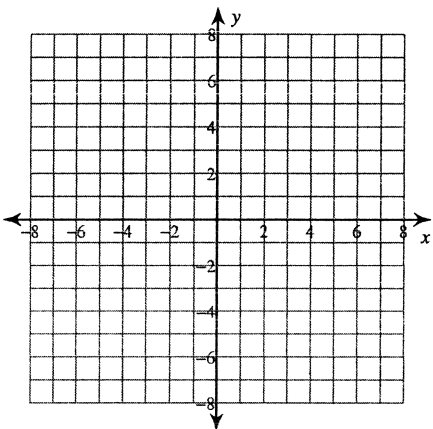
32) $f(x) = x^4 - 3x^2 + 3x - 1$

33) $f(x) = -x^4 + 4x^3 - 4x^2 - 1$

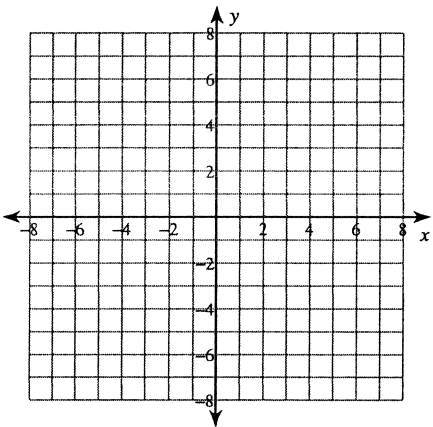


34) $f(x) = -x^5 + 4x^3 - x - 1$

35) $f(x) = x^5 - 4x^3 + 5x + 4$



36) $f(x) = x^5 - 4x^3 + x$



Name each polynomial by degree and number of terms.

37) $x^8 + 4 + 6x^3$

38) $-4 + 5m$

39) $6n^4 + 10 - 9n^2 - 9n^3$

Practice Worksheet

The Rational Zero Theorem

State all possible rational zeros for each function.

1. $f(x) = x^3 - 5x^2 + 2x + 12$

2. $f(x) = x^4 - 8x^3 + 7x - 4$

3. $f(x) = 5x^4 - 2x - 4$

4. $f(x) = 3x^5 - 7x^2 + x + 6$

Find all the rational zeros for each function.

5. $f(x) = x^3 + 3x^2 - 6x - 8$

6. $f(x) = x^3 + 7x^2 + 7x - 15$

7. $f(x) = x^3 - 9x^2 + 27x - 27$

8. $f(x) = x^3 - x^2 - 8x + 12$

9. $f(x) = x^4 - 3x^3 - 11x^2 + 3x + 10$

10. $f(x) = x^4 - 4x^3 - 7x^2 + 34x - 24$

11. $f(x) = x^4 - 2x^3 - 4x^2 + 11x - 6$

12. $f(x) = x^3 + 4x^2 - 2x + 15$

Find all zeros of each function.

13. $f(x) = 3x^3 - 4x^2 - 17x + 6$

14. $f(x) = 4x^3 - 12x^2 - x + 3$

15. $f(x) = 18x^3 + 9x^2 - 2x - 1$

16. $f(x) = 2x^3 + 3x^2 + 5x + 2$

17. $f(x) = 2x^4 + 7x^3 - 2x^2 - 19x - 12$

18. $f(x) = x^4 - 4x^3 + x^2 + 16x - 20$

Practice 6-5**Theorems about Roots of Polynomial Equations**

A polynomial equation with rational coefficients has the given roots.

Find two additional roots.

1. $2 + 3i$ and $\sqrt{7}$

2. $3 - \sqrt{2}$ and $1 + \sqrt{3}$

3. $-4i$ and $6 - i$

4. $5 - \sqrt{6}$ and $-2 + \sqrt{10}$

Find a fourth-degree polynomial equation with integer coefficients that has the given numbers as roots.

5. $2i$ and $4 - i$

6. $\sqrt{2}$ and $2 - \sqrt{3}$

7. $3i$ and $\sqrt{6}$

8. $2 + i$ and $1 - \sqrt{5}$

Find the roots of each polynomial equation.

9. $x^3 - 5x^2 + 2x + 8 = 0$

10. $x^3 + x^2 - 17x + 15 = 0$

11. $2x^3 + 13x^2 + 17x - 12 = 0$

12. $x^3 - x^2 - 34x - 56 = 0$

13. $x^3 - 18x + 27 = 0$

14. $x^4 - 5x^2 + 4 = 0$

15. $x^3 - 6x^2 + 13x - 10 = 0$

16. $x^3 - 5x^2 + 4x + 10 = 0$

17. $x^3 - 5x^2 + 17x - 13 = 0$

18. $x^3 + x + 10 = 0$

19. $x^3 - 5x^2 - x + 5 = 0$

20. $x^3 - 12x + 16 = 0$

21. $x^3 - 2x^2 - 5x + 6 = 0$

22. $x^3 - 8x^2 - 200 = 0$

23. $x^3 + x^2 - 5x + 3 = 0$

24. $4x^3 - 12x^2 - x + 3 = 0$

25. $x^3 + x^2 - 7x + 2 = 0$

26. $12x^3 + 31x^2 - 17x - 6 = 0$

Use the Rational Root Theorem to list all possible rational roots for each polynomial equation. Then find any actual rational roots.

27. $x^3 + 5x^2 - 2x - 15 = 0$

28. $36x^3 + 144x^2 - x - 4 = 0$

29. $2x^3 + 5x^2 + 4x + 1 = 0$

30. $12x^4 + 14x^3 - 5x^2 - 14x - 4 = 0$

31. $5x^3 - 11x^2 + 7x - 1 = 0$

32. $x^3 + 81x^2 - 49x - 49 = 0$

Find a third-degree polynomial equation with rational coefficients that has the given numbers as roots.

33. $3, 2 - i$

34. $5, 2i$

35. $-1, 3 + i$

36. $-7, i$

37. $-4, 4i$

38. $6, 3 - 2i$

When finding the roots of a quartic, use the Rational Root Theorem to find TWO real rational roots, and then do synthetic division twice to find the remaining roots.

Examples: Find all roots for the following polynomial equations

A) $9x^4 + 3x^3 - 30x^2 + 6x + 12 = 0$

B) $x^4 - 2x^3 - 5x^2 + 8x + 4 = 0$

You Try: Find all the roots of $x^4 - 2x^2 - 16x - 15 = 0$

"What do skeletons say before dinner?"

Find all of the roots for each polynomial. Cross out the letter that matches with your answer. The remaining letters will allow you to figure out the joke.

1. $y = x^3 - 6x^2 + 11x - 6$

O: 3, 4, 5

2. $y = x^3 + 5x^2 - 4x - 20$

W: 0, -1, -2, -3

Y: -7, 5, $\frac{1}{3}$

3. $y = x^3 - x^2 - 14x + 24$

D: 2, -2, -5

B: 2, -3, 4

4. $y = 3x^3 + 5x^2 - 107x + 35$

P: 0, 3, 5, -8

C: 1, 2, 3

5. $y = 6x^3 - 11x^2 - 24x + 9$

T: 2, 3, $\frac{1}{4}$

S: -3, -1, 1, 2

6. $y = x^3 + 3x^2 + 4x + 12$

U: 3, $\frac{1}{3}$, $-\frac{3}{2}$

E: 6, -2, 0, 1

7. $y = x^4 + 6x^3 + 11x^2 + 6x$

O: -5, 5, 6

Q: $-2i$, $2i$, -3

N: $4i$, $-4i$, 3

8. $y = x^4 + x^3 - 7x^2 - x + 6$

M: 3, -4, 2

W Y B M O N Q W E A S P C P E T D I U T E

No hints, find all of the roots

Joke #37

Practice 6-2**Polynomials and Linear Factors**

For each function, determine the zeros. State the multiplicity of any multiple zeros.

1. $y = (x - 5)^3$

2. $y = x(x - 8)^2$

3. $y = (x - 2)(x + 7)^3$

4. $f(x) = x^4 - 8x^3 + 16x^2$

5. $f(x) = 9x^3 - 81x$

6. $y = (2x + 5)(x - 3)^2$

Write each function in standard form.

7. $y = (x - 5)(x + 5)(2x - 1)$

8. $y = (2x + 1)(x - 3)(5 - x)$

9. A rectangular box is 24 in. long, 12 in. wide, and 18 in. high. If each dimension is increased by x in., write a polynomial function in standard form modeling the volume V of the box.

Write a polynomial function in standard form with the given zeros.

10. $-1, 3, 4$

11. $1, 1, 2$

12. $-3, 0, 0, 5$

13. -2 multiplicity 3

Write each expression as a polynomial in standard form.

14. $x(x - 1)^2$

15. $(x + 3)^2(x + 1)$

16. $(x + 4)(2x - 5)(x + 5)^2$

Write each function in factored form. Check by multiplication.

17. $y = 2x^3 + 10x^2 + 12x$

18. $y = x^4 - x^3 - 6x^2$

19. $y = -3x^3 + 18x^2 - 27x$

Find the zeros of each function. Then graph the function.

20. $y = (x + 1)(x - 1)(x - 3)$

21. $y = (x + 2)(x - 3)$

22. $y = x(x - 2)(x + 5)$

Find the relative maximum, relative minimum, and zeros of each function.

23. $f(x) = x^3 - 7x^2 + 10x$

24. $f(x) = x^3 - x^2 - 9x + 9$

Write each polynomial in factored form. Check by multiplication.

25. $x^3 - 6x^2 - 16x$

26. $x^3 + 7x^2 + 12x$

27. $x^3 - 8x^2 + 15x$

28. A rectangular box has a square base. The combined length of a side of the square base, and the height is 20 in. Let x be the length of a side of the base of the box.

- Write a polynomial function in factored form modeling the volume V of the box.
- What is the maximum possible volume of the box?

Factors and Zeros

Find all zeros.

1) $f(x) = (2x - 1)(x - 5)$

2) $f(x) = (x - 3)(3x + 1)(x + 1)$

3) $f(x) = (2x + 1)(x + 1)(x - 1)$

4) $f(x) = x(5x - 2)(x^2 + 1)$

5) $f(x) = x(x + 2)(x - 2)(3x^2 - 4)$

6) $f(x) = (2x - 1)(x^2 + 3)(2x^2 - 5)$

7) $f(x) = x(2x - 1)(x - 1)(x + 1)$

8) $f(x) = (2x + 5)(x^2 - 2x - 5)$

Name _____

Date _____ Period _____

11) 5, -1, 0

12) -3, $-\frac{1}{3}$, 5

13) $\frac{5}{3}$, 1, -1

14) 2, $\frac{5}{3}$, -5

Find all zeros by factoring each function.

15) $f(x) = x^3 - 2x^2 + x$

16) $f(x) = x^3 + 8$

17) $f(x) = x^4 - x^2 - 30$

18) $f(x) = x^4 + x^2 - 12$

Write a polynomial function of least degree with integral coefficients that has the given zeros.

9) 3, 2, -2

10) 3, 1, -2, -4

19) $f(x) = x^6 - 64$

20) $f(x) = x^6 + 2x^3 + 1$